

Recognise and explain the connection between multiplication and division as inverse operations and use this to develop families of number facts

Learning intention

- recognise and explain the connection between multiplication and division as inverse operations and use this to develop families of number facts.

Teach and model

- using arrays, multiplication tables, and physical and virtual materials to develop families of facts.
- using materials, diagrams or arrays to recognise and explain the inverse relationship between multiplication and division.
- using materials or diagrams to develop and explain division strategies, such as halving, using the inverse relationship to turn division into a multiplication.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.